

# Joshua Mark

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## EDUCATION

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### UNC Chapel Hill

**Expected: May 2027**

*B.S. Geospatial Data Science, B.S. Information Science, Minor Music*

*Chapel Hill, NC*

- **Cumulative GPA:** 3.991/4.0 | **Dean's List:** all semesters.
- **Relevant Coursework:** Applied Spatial Data Science, GIS Programming, Cartography & Geovisualization, Intro to Remote Sensing.

## TECHNICAL SKILLS

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**GIS & Spatial Analysis:** ArcGIS Pro, ArcGIS Online, QGIS, GeoDa.

**Programming & Data Analysis:** Python (ArcPy, pandas); R (sf, tmap, tidyverse, tidycensus, terra); SQL.

**Spatial & Statistical Methods:** Exploratory data analysis, Regression (OLS + spatial) modeling, hypothesis testing.

**Visualization & Development:** Tableau, HTML, CSS, PHP, Git/GitHub.

## PROJECTS

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### [NC Triangle Walkability Analysis](#)

- Built a 4-script pipeline in R, merging EPA Walkability Index with HUD LMI dataset to find a relationship between walkability and income within NC Triangle.
- Ran an OLS regression, diagnosed spatial autocorrelation, and used Lagrange Multiplier (LM) test to determine best fit. Ran and interpreted a spatial lag and SARMA model to find significance.
- Found significance in the opposite direction of initial hypothesis. Visualized results using a stylized bivariate choropleth map.

### Data Centers and Electricity Tool

- Procured and cleaned messy datasets: electricity rates by ZIP code and crowdsourced data center locations.
- Scripted an ArcPy tool to take a user-input ZIP code and calculate relevant statistics: distance to nearest data center, local area-weighted average electricity price, and nearest data center average electricity rate.
- Performed a time-series analysis, comparing change in electricity rate over 10 years using percentage change, and integrated it into data center comparison.

### [Residential Access to Grocery Stores](#)

- Used QGIS to create an ORS network isochrone analysis, calculating areas over a 5-minute drive or 20-minute walk from grocery stores in Chapel Hill
- Clipped data using a Chapel Hill jurisdictional boundary and intersected with an externally cleaned neighborhood polygon file to accurately represent residential area.
- Designed a professional cartographic map layout, emphasizing areas that lack equal access to grocery stores.

### Comparative Segregation Analysis of Detroit vs. Tampa

- Conducted a comparative case study analysis on Index of Concentrations at the Extremes (ICE) values across Detroit and Tampa, integrating local stories and city history to hypothesize spatial processes.
- Ran an exploratory spatial analysis by comparing ICE descriptive statistics and mapping values between cities, then performed a global and local structure clustering analysis on each.
- Created a multivariate spatial regression to measure the impact of historical redlining, contemporary home values, and environmental health indicators. Interpreted values and offered explanations for nonstationarity.

## RESEARCH EXPERIENCE

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### UNC Chapel Hill, Human-AI Interactions in Knowledge Work

**2024 – 2025**

*Undergrad Research Assistant*

*Chapel Hill, NC*

- Key role in shaping recruitment and candidate screening process.
- Managed Qualtrics survey entries and distributed programmed emails to 54 participants over 3 cohorts.
- Conducted semi-structured, conversational interviews with 10 participants and synthesized findings.